https://colab.research.google.com/drive/11GCNfPIG1zt2SKzBWZEivKJsrgqaApRk?usp=sharing

9.21 => Given: 8.3, 10.6, 9.7, 8.8, 10.2, 9.4
$n = 6$

a.) $8.3 + 10.6 + 9.7 + 8.8 + 10.2 + 9.4 = 57.0$
$\Rightarrow \bar{x} = \frac{57.0}{6} = 9.5 \Rightarrow \boxed{\mu \approx 9.5\,kg}$

b.) $\Rightarrow 1.44 + 1.21 + 0.04 + 0.49 + 0.01 = 3.68$
$\Rightarrow s^2 = \frac{3.68}{6-1} \Rightarrow s^2 = \frac{3.68}{5} \Rightarrow s^2 = 0.736 \Rightarrow \boxed{\sigma^2 \approx 0.736}$

c.) $\Rightarrow s = \sqrt{s^2} \Rightarrow s = \sqrt{0.736} \Rightarrow s = 0.858$
$\Rightarrow \sigma = \sqrt{\frac{3.68}{6}} \Rightarrow \sigma = \sqrt{0.6133} \Rightarrow \boxed{\sigma = 0.783}$

9.24.) 95% confidence level: $Z_{\alpha/2} = 1.96$
$E = 1.96(0.0942)$
$E = 0.1846$
$\Rightarrow$ lower: $11.09 - 0.1846 = 10.9054$
$\Rightarrow$ upper: $11.09 + 0.1846 = 11.2746$
$\boxed{= (10.905,\ 11.275)}$

b.) 99% confidence level $\rightarrow Z_{\alpha/2} = 2.576$
$E = 2.576(0.0942)$
$E = 0.2427$
lower: $11.09 - 0.2427 = 10.8473$
upper: $11.09 - 0.2427 = 11.3327$
$\boxed{= (10.847,\ 11.333)}$

9.33 -> 95% confidence level -> $Z_{\alpha/2} = 1.96$
$E = 1.96(0.0453)$
$E = 0.0888$
lower: $1.07 - 0.0888 = 0.9812$
upper: $1.07 + 0.0888 = 1.1588$
$\boxed{= (0.981,\ 1.159)}$

9.34.) males $n_1 = 50$, females $n_2 = 50$

$\Rightarrow \bar{x}_m = 69.74$

$\bar{x}_f = 78.44$

$s_m = 16.24$

$s_f = 14.90$

$\Rightarrow 69.74 - 78.44 = -8.70$

$$\Rightarrow SE = \sqrt{\frac{16.24^2}{50} + \frac{14.90^2}{50}} = SE = 3.116$$

$\Rightarrow 85\%$ CI $= \alpha = 0.15$

$\alpha/2 = 0.075$ $= t_{0.925, 97} \approx 1.451$

$\Rightarrow ME = 1.451(3.116)$

$ME = 4.52$

Lower: $-8.70 - 4.52 = \boxed{-13.22}$

Upper: $-8.70 + 4.52 = \boxed{-4.18}$